

Website Traffic Analysis Project

1. Business Insights

The Website Traffic Analysis project focuses on analyzing website traffic, user engagement, conversion rates, and page performance using Python. The main objective is to understand how users interact with a website and identify patterns in traffic sources and page performance.

The analysis was performed using Pandas, Matplotlib, and Seaborn to process the dataset and create visualizations.

1.1 Traffic Source Analysis

Organic Search generated the highest number of sessions, with 25,932 sessions, followed by Direct traffic with 19,582 sessions and Paid Search with 17,148 sessions.

Insight: Organic Search was the largest traffic source in the dataset, indicating that search engines contributed a substantial share of website visits.

1.2 Bounce Rate Analysis

Paid Social recorded the highest bounce rate at 57.4%, followed by Organic Social at 52.9%. Email recorded the lowest bounce rate at 26.6%.

Insight: Social media traffic had higher bounce rates, while Email traffic had a comparatively lower bounce rate.

1.3 Conversion Rate Analysis

Email achieved the highest conversion rate at 8.94%, followed by Referral at 6.33% and Paid Search at 6.23%.

Insight: Email was the traffic source with the highest conversion rate in the dataset.

1.4 Page Performance Analysis

The Home page received the highest number of pageviews, with 48,568 pageviews, followed by Products with 43,800 pageviews and Product A with 31,699 pageviews.

Insight: The Home and Products pages attracted the highest number of pageviews.

1.5 Website Traffic Trend

The daily website traffic chart displays changes in sessions over time. It helps identify fluctuations in traffic and observe daily patterns.

Insight: Monitoring daily sessions can help identify traffic changes and support website performance analysis.

2. Recommendations

1. Improve Organic Search: Optimize website content and search engine visibility to maintain and improve organic traffic.
2. Optimize Social Media Traffic: Review social media campaigns and landing pages to understand the high bounce rates.

3. **Strengthen Email Marketing:** Analyze successful email campaigns and improve content and calls to action to support conversions.
4. **Improve High-Traffic Pages:** Enhance the Home and Products pages with clear navigation and relevant content.
5. **Monitor Website Performance:** Regularly track sessions, bounce rate, conversion rate, and pageviews.

3. Conclusion

The Website Traffic Analysis project demonstrates how Python-based data analysis and visualization can be used to understand website traffic, user engagement, conversion performance, and page popularity.

The dashboard presents key performance indicators and charts that help identify traffic sources, compare conversion rates, and evaluate page performance. These insights can support data-driven decisions for improving website engagement and conversion performance.

4. Dataset Note

The dataset used in this project is synthetic and was created for practice and demonstration purposes. The findings represent simulated data and should not be interpreted as actual website analytics.